

Sep 15, 2024 (Due: 08:00 Sep 22, 2025)

1. Let $\hat{x}$ be an approximation to x . In practice it is often much easier to estimate $\tilde{E}_{\text{rel}}(\hat{x}) = |x - \hat{x}|/|\hat{x}|$ compared to $E_{\text{rel}}(\hat{x}) = |x - \hat{x}|/|x|$. What is the relationship between E_{rel} and $\tilde{E}_{\text{rel}}$?
2. You are given $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$, both already stored in floating-point format. Show that there exists a “small” matrix $E \in \mathbb{R}^{m \times n}$ such that $\text{fl}(Ax) = (A + E)x$. Try to bound the entries of E as tight as you can. You may assume that there is no overflow or (gradual) underflow in the calculation.
3. Let A be a nonsingular banded matrix with half-bandwidth b (i.e., $a_{ij} = 0$ if $|i - j| > b$). Suppose that the LU factorization of A (without pivoting) is $A = LU$. Show that both L and U are banded matrices with half-bandwidth b .
4. Find the exact LU factorization of the $n \times n$ matrix

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 1 \\ -1 & 1 & 0 & \cdots & 0 & 1 \\ -1 & -1 & 1 & \cdots & 0 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -1 & -1 & \cdots & 1 & 1 \\ -1 & -1 & -1 & \cdots & -1 & 1 \end{bmatrix}.$$

5. Find the exact Cholesky factorization of the $n \times n$ *Pascal matrix*

$$P_n = \begin{bmatrix} 1 & 1 & 1 & 1 & \cdots \\ 1 & 2 & 3 & 4 & \cdots \\ 1 & 3 & 6 & 10 & \cdots \\ 1 & 4 & 10 & 20 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \in \mathbb{R}^{n \times n}.$$

Compare the exact solution with the computed one. What do you observe?

Note: the (i, j) -entry of the Pascal matrix is

$$\binom{i+j-2}{i-1} = \frac{(i+j-2)!}{(i-1)!(j-1)!}.$$

6. (H) Download OpenBLAS and LAPACK from the internet and compile them on your own computer. Run the test programs to ensure that you have successfully compiled the packages. You may write down the key steps and take some screenshots.

7. (optional) Suppose that you have a BLAS library with an inefficient implementation of **STRSM**. Try to design an efficient **STRSM** subroutine by making use of **SGEMM**. You do not have to really implement it. Just describe your algorithm.

Hint: Search online if you do not know what **STRSM** and **SGEMM** stand for. Useful information can be found at, e.g., <http://www.netlib.org/blas/>.